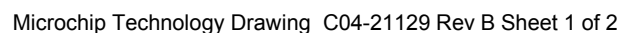
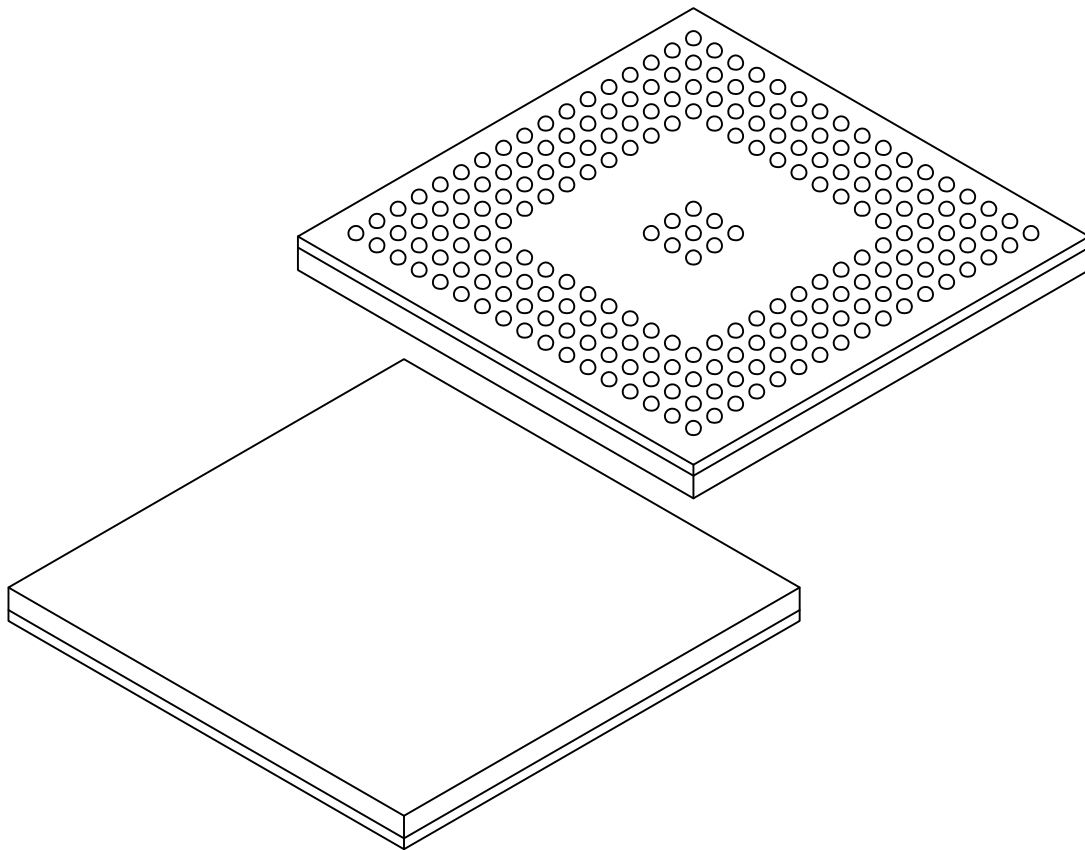


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**217-Ball Low Profile Fine Pitch Ball Grid Array (ATB) - 15x15x1.4 mm Body [LFBGA]
Atmel Legacy Global Package Code CGF**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



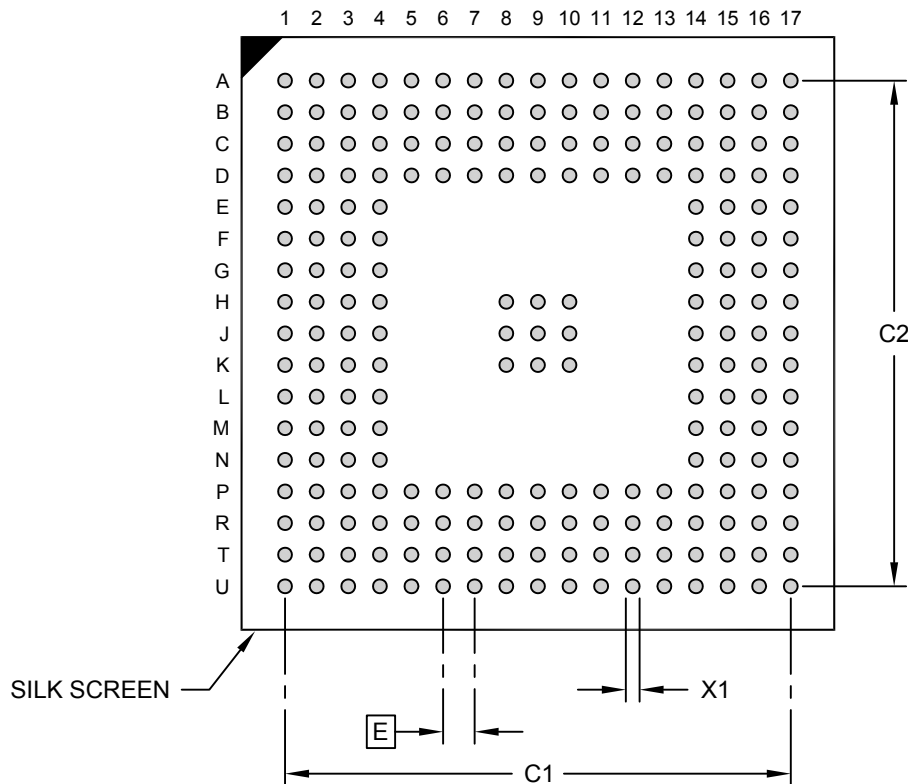
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	217		
Pitch	e	0.80 BSC		
Overall Height	A	1.20	1.30	1.40
Ball Height	A1	0.25	0.30	0.35
Substrate Thickness	A2	0.32	0.36	0.40
Mold Cap Height	A3	0.48	0.53	0.58
Overall Length	D	15.00 BSC		
Overall Width	E	15.00 BSC		
Terminal Width	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

217-Ball Low Profile Fine Pitch Ball Grid Array (ATB) - 15x15x1.4 mm Body [LFBGA] Atmel Legacy Global Package Code CGF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		12.80	
Contact Pad Spacing	C2		12.80	
Contact Pad Diameter (X217)	X1			0.35

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23129 Rev B